Jinsong Zhang

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Education

University of California, Santa Barbara — Santa Barbara, USA
M.S. in Computer Engineering; GPA: 3.78/4.0 — *Sep 2024 – Dec 2025*

University of Liverpool — Liverpool, UK
B.Eng. in Electrical and Electronic Engineering; GPA: 3.7/4.0 — *Sep 2022 – Jun 2024*

Xi'an Jiaotong-Liverpool University — Suzhou, China
B.Eng. in Electronic Science and Technology; GPA: 3.3/4.0 — *Sep 2020 – Jun 2022*

Publications

J. Zhang, M. Li, J. Tian, J. Lu, Z. Zhang. "Beyond MACs: Deployment-Constrained Design Space Exploration for Tensorized Neural Networks" Available at *arXiv preprint arXiv:2511.17971*, 2025.

Research Projects

Beyond MACs: Deployment-Constrained DSE for Tensorized Neural Networks Jan 2025 – Nov 2025
Research Assistant, University of California, Santa Barbara — *Santa Barbara, USA*

- Proposed a deployment-constrained Design Space Exploration (DSE) framework that jointly optimizes tensor contraction paths, PE-array partitioning, and dataflow mappings for efficient TNN deployment.
- Developed a global latency-driven search algorithm with MAC-guided path pruning to identify latency-optimal configurations under hardware constraints, bridging algorithmic compression and hardware efficiency.
- Designed and implemented a parameterizable FPGA accelerator featuring a streaming Tensor Train (TT) contraction kernel and configurable INT8 systolic GEMM engine using Vitis HLS on Xilinx VU9P.
- Achieved 3.28x–4.00x inference and 3.42x–3.85x training speedups over INT8 dense baselines under matched PE budgets across ResNet-18 and ViT-Ti/4 benchmarks.

Scalable Spiking Neural Networks via Reward Broadcasting Jan 2025 – Mar 2025
Graduate Researcher, University of California, Santa Barbara — *Santa Barbara, USA*

- Designed a scalable Spiking Neural Network (SNN) learning framework integrating reinforcement learning (RL) into Brain-Inspired Reward Broadcasting (BIRB) to address deep credit assignment and catastrophic forgetting.
- Implemented a hybrid learning scheme combining hierarchical global-to-local reward broadcasting with surrogate gradient descent, overcoming vanishing reward issues without the high memory overhead of BPTT.
- Formulated an SNN-Q learning algorithm utilizing multi-step spike-feature extraction, temporal-difference updates, and experience replay to achieve robust sequential continual learning without task-specific memory buffers.
- Achieved SOTA SNN accuracy of 96.88% on MNIST and 83.64% on Fashion-MNIST, while reducing training convergence epochs by up to 66%.

Custom 32-bit 5-Stage Pipelined RISC-V Processor Sep 2024 – Dec 2024
VLSI Project, University of California, Santa Barbara — *Santa Barbara, USA*

- Designed a 32-bit 5-stage pipelined RISC-V processor in Verilog, implementing hazard detection, data forwarding, and stall mechanisms to eliminate pipeline data/control hazards.
- Constructed the complete transistor-level processor datapath from scratch in Cadence Virtuoso using the NCSU 45nm FreePDK, targeting robust 50 MHz operation across all pipeline stages.
- Engineered test stimuli for a 10-instruction benchmark with zero-state initialization, performing transient simulations in Cadence to cross-verify cycle-by-cycle behavioral equivalence between RTL and transistor schematics.

Research & Professional Experience

Visiting Research Assistant – Autonomous Robotics & Perception May 2023 – Sep 2023
Westlake University — *Hangzhou, China*

- **Robot Swarm Edge Vision:** Built a distributed real-time perception pipeline deploying custom YOLOv5s models over ROS on edge embedded platforms; curated and annotated domain-specific robot swarm datasets.
- **Inference Optimization:** Refactored deep learning inference pipelines and ROS communication nodes from Python to C++, dramatically reducing edge latency and increasing control-loop throughput.
- **UAV Terrain Sensing Payload:** Engineered an airborne terrain-monitoring module integrating an IMU and multi-beam laser distance sensors, establishing high-throughput, low-latency UART telemetry.
- **Embedded Signal Filtering:** Implemented dynamic moving-average filtering on microcontrollers to suppress motor vibration noise, providing robust real-time slope estimation for terrain-adaptive flight control.

Skills

Programming: C/C++, Verilog, Python, PyTorch, MATLAB, Assembly
Software: Vivado, Quartus, Vitis HLS, Cadence, VS Code, PyCharm, Multisim